Section 3–1 The Inverter:

1. The input waveform shown in Figure 3–76 is applied to an inverter. Draw the timing diagram of the output waveform in proper relation to the input.

FIGURE 3–76

2. A combination of inverters is shown in Figure 3–77. If a High is applied to point *A*, determine the net output at points *E* and *F*.

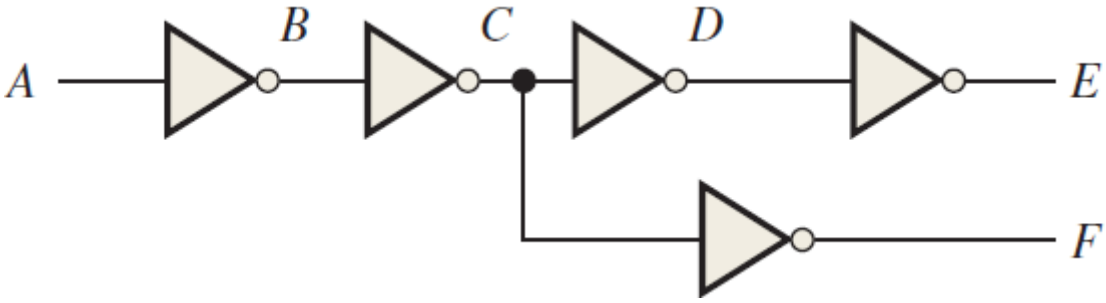


FIGURE 3–77

Section 3–2 The AND Gate:

5. Determine the output, *X*, for a 2-input AND gate with the input waveforms shown in Figure 3–78. Show the proper relationship of output to inputs with a timing diagram.

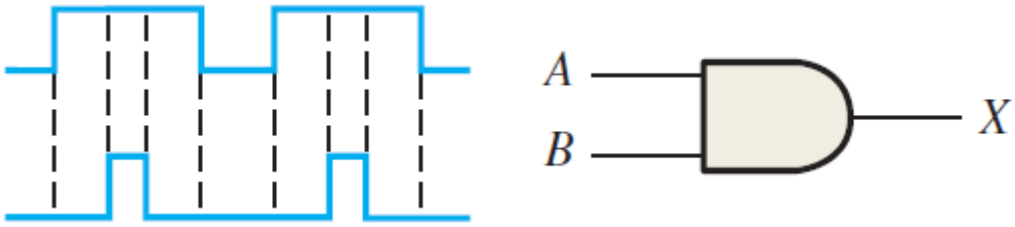


FIGURE 3–78

8. The input waveforms applied to a 4-input AND gate are as indicated in Figure 3–81. Show the output waveform in proper relation to the inputs with a timing diagram.

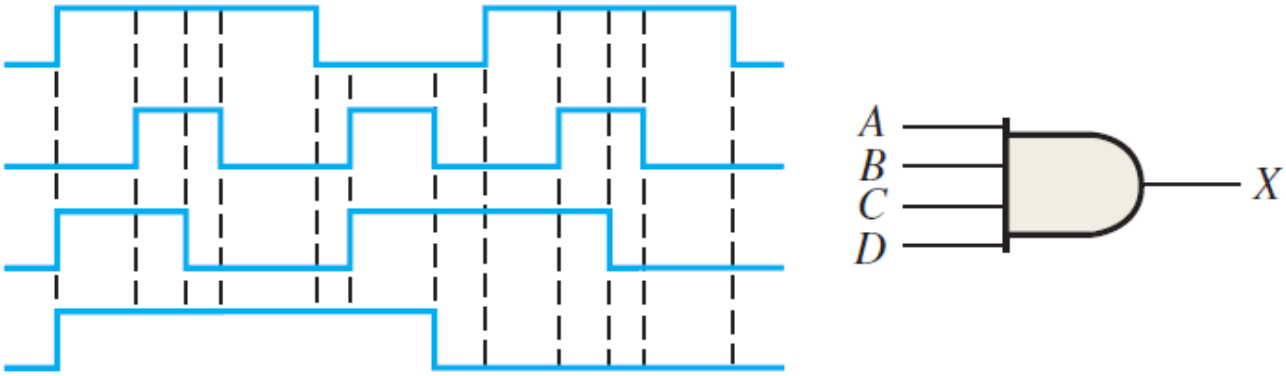


FIGURE 3–81

Section 3–3 The OR Gate:

9. Determine the output for a 2-input OR gate when the input waveforms are as in Figure 3–79 and draw a timing diagram.

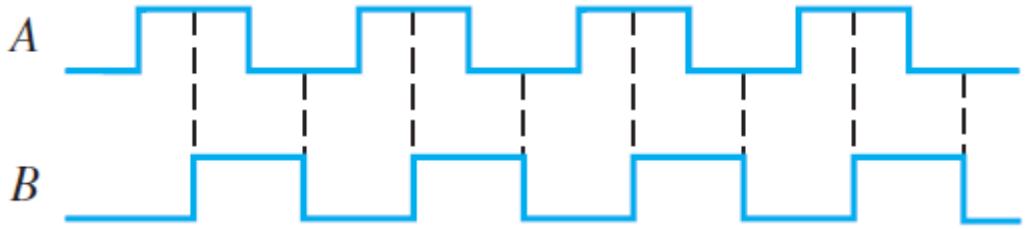


FIGURE 3–79

12. For the waveforms given in Figure 3–82, *A* and *B* are ANDed with output *F*, *D* and *E* are ANDed with output *G*, and *C*, *F*, and *G* are ORed. Draw the net output waveform.

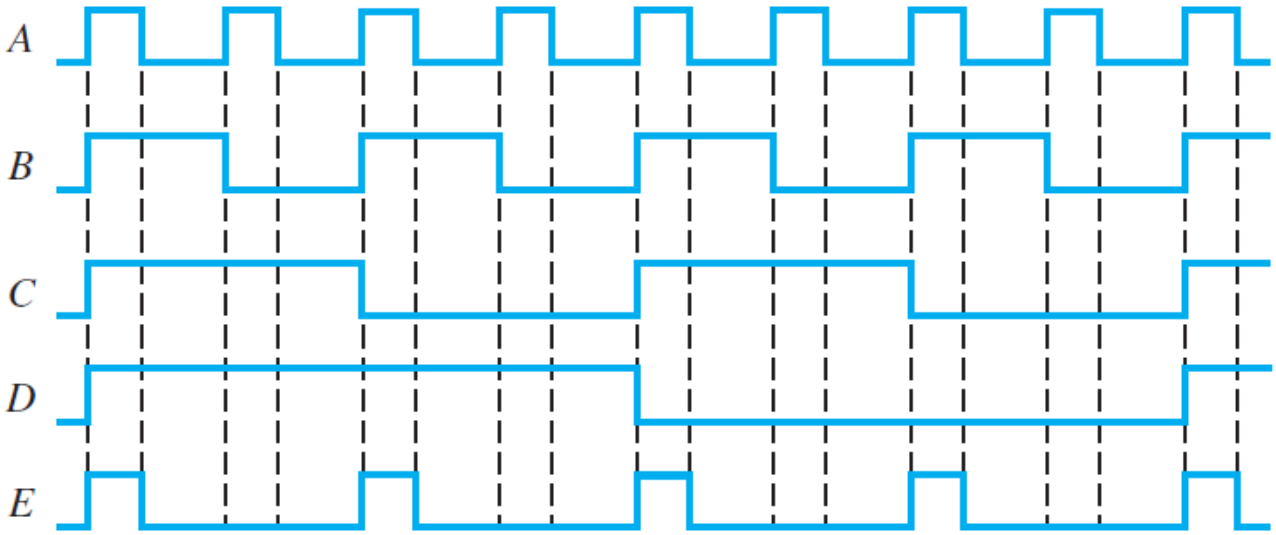


FIGURE 3–82

Section 3–4 The NAND Gate:

17. Determine the output waveform in Figure 3–85.

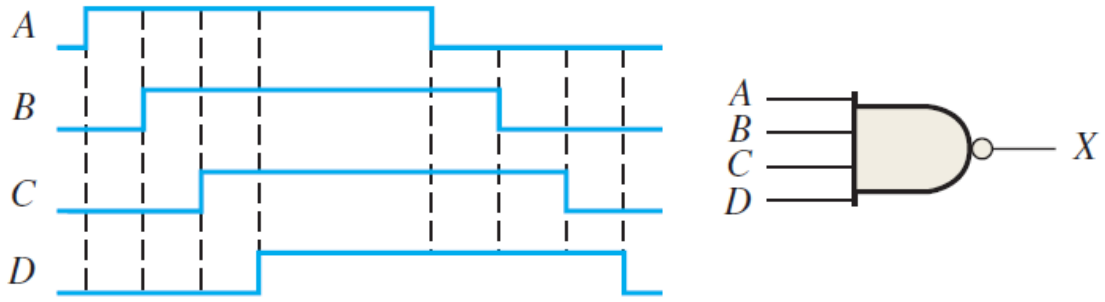


FIGURE 3–85

Section 3–5 The NOR Gate:

22. The NAND and the negative-OR symbols represent equivalent operations, but they are functionally different. For the NOR symbol, look for at least one HIGH on the inputs to give a LOW on the output. For the negative-AND, look for two LOWs on the inputs to give a HIGH output. Using these two functional points of view, show that both gates in Figure 3–88 will produce the same output for the given inputs.

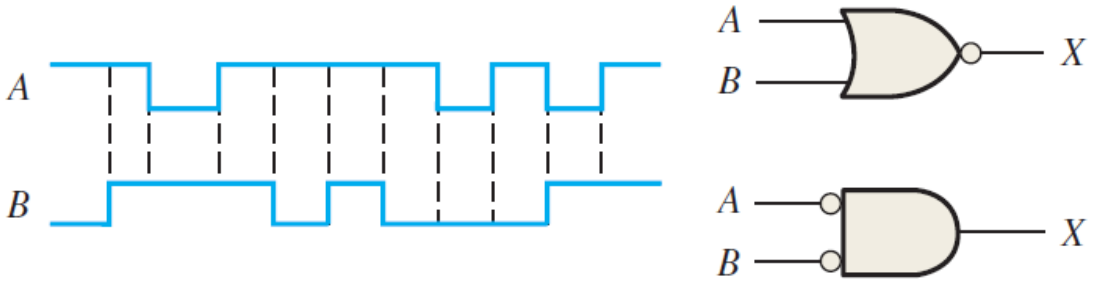


FIGURE 3–88

Section 3–6 The Exclusive-OR and Exclusive-NOR Gates:

25. For the set of input waveforms in Figure 3–83, determine the output for an exclusive-NOR gate and draw the timing diagram.

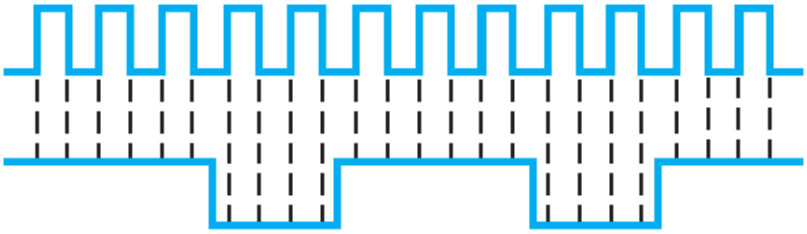
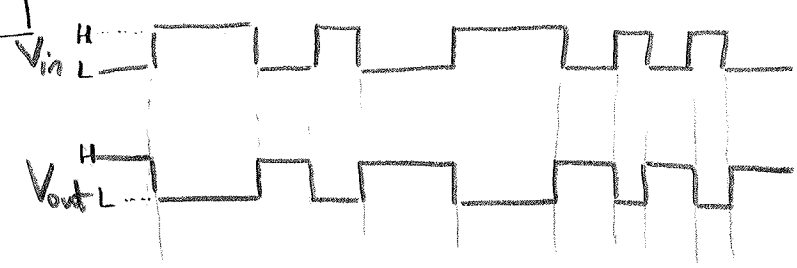
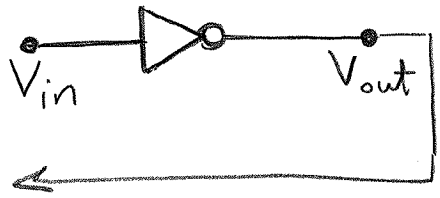


FIGURE 3–83

26. Determine the output of an exclusive-OR gate for the inputs shown in Figure 3–79 and draw a timing diagram.

1





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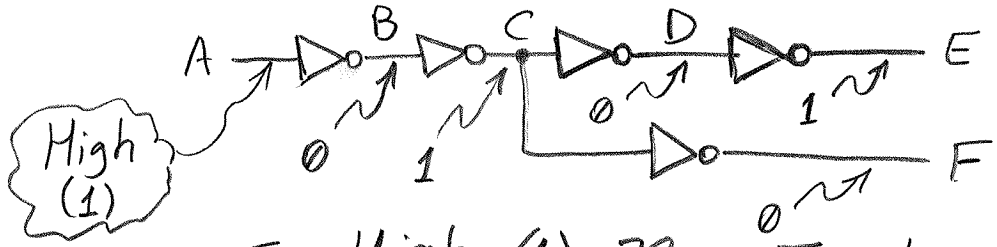
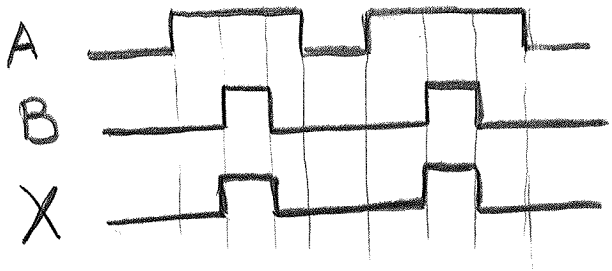
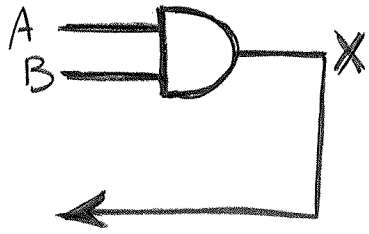


E = High (1) & F = Low (0)

5





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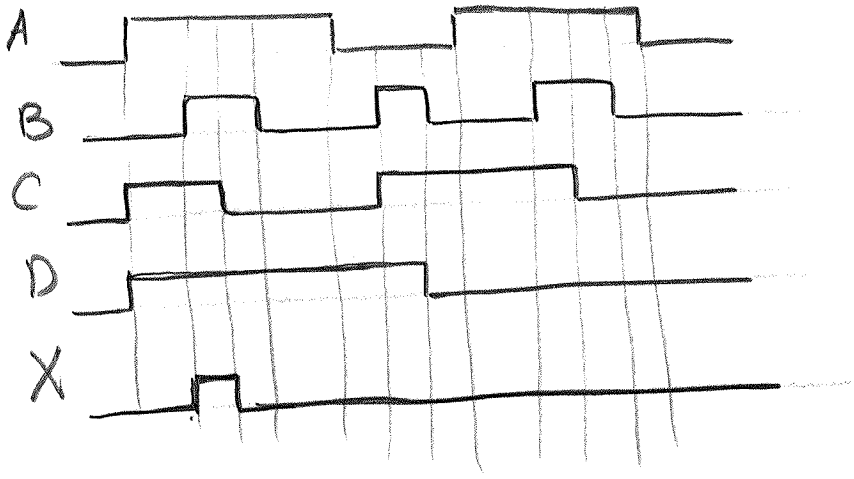

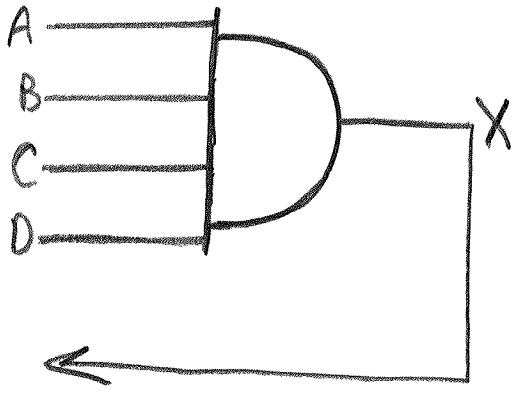

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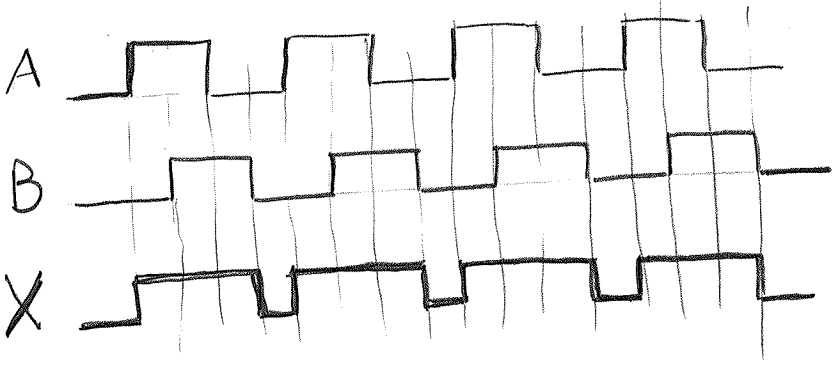

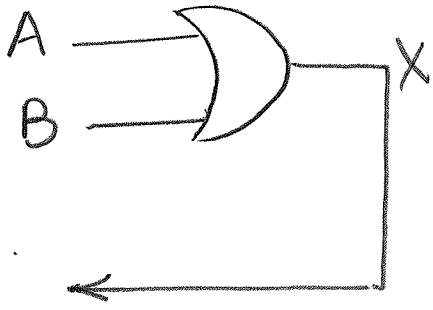

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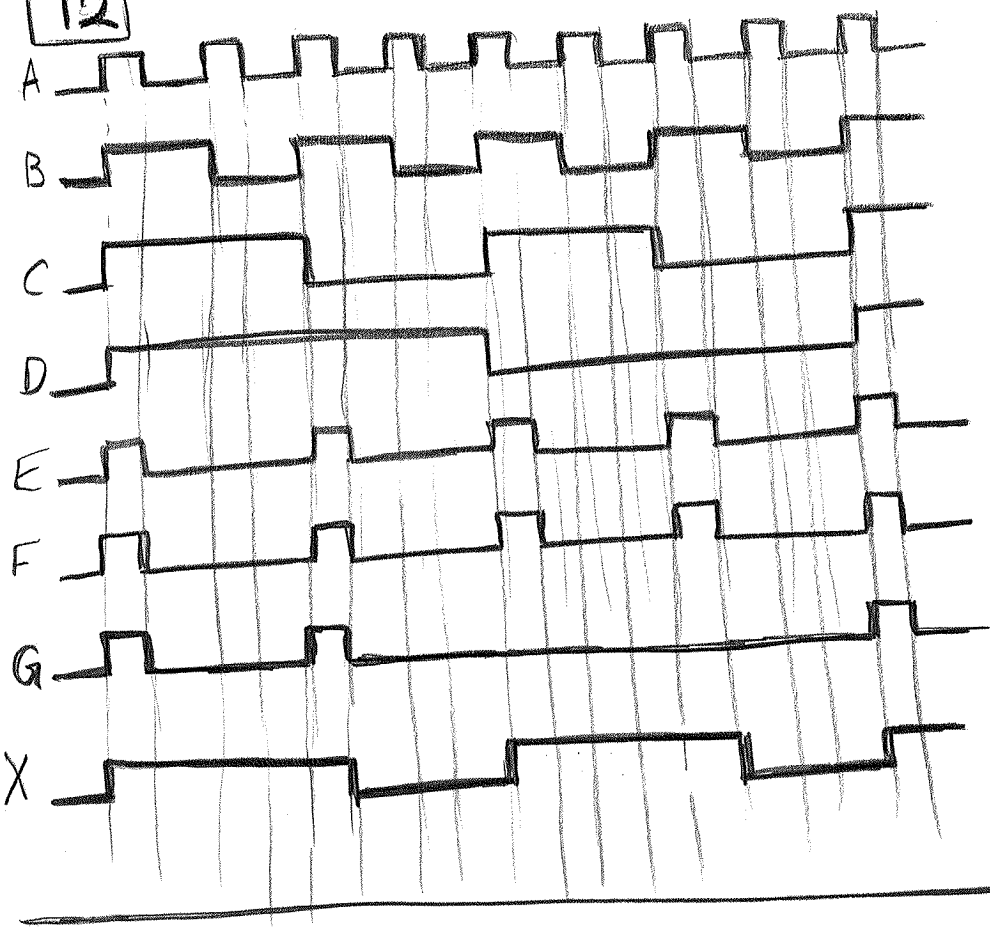
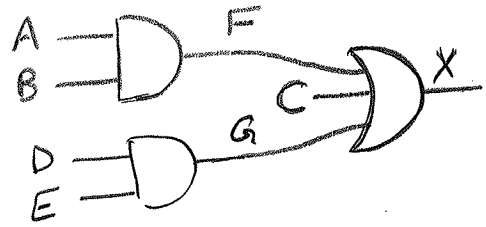
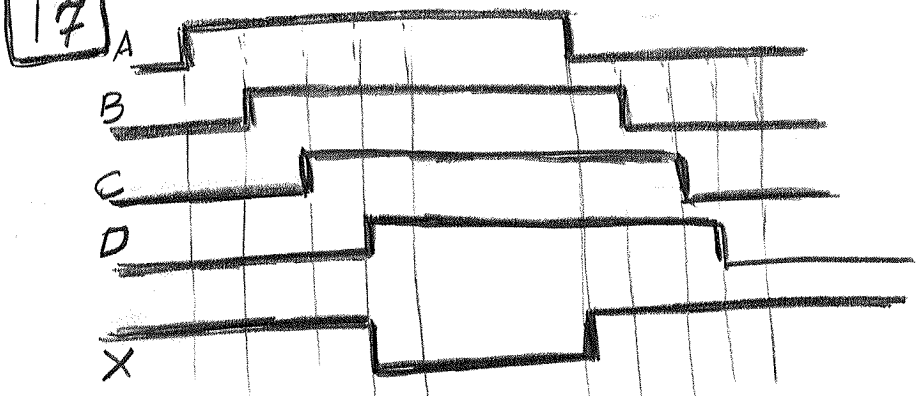
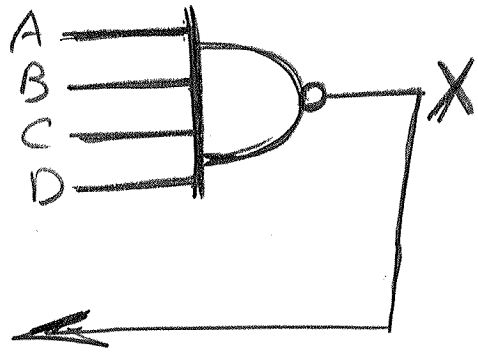




ملحوظة: تم تعديل المطلوب
إلى الدائرة المنطقية هذه
بدلاً من
و

17





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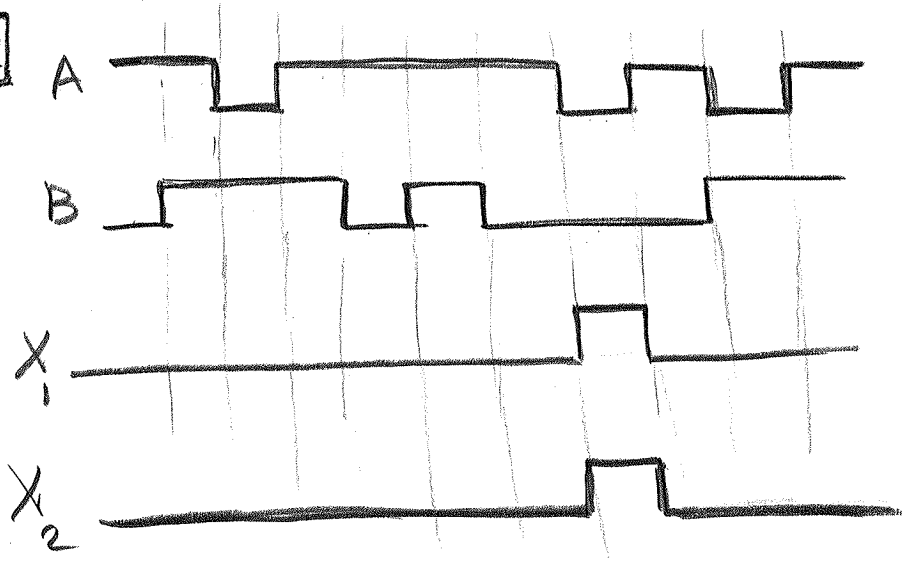

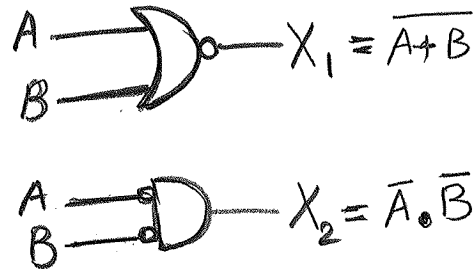

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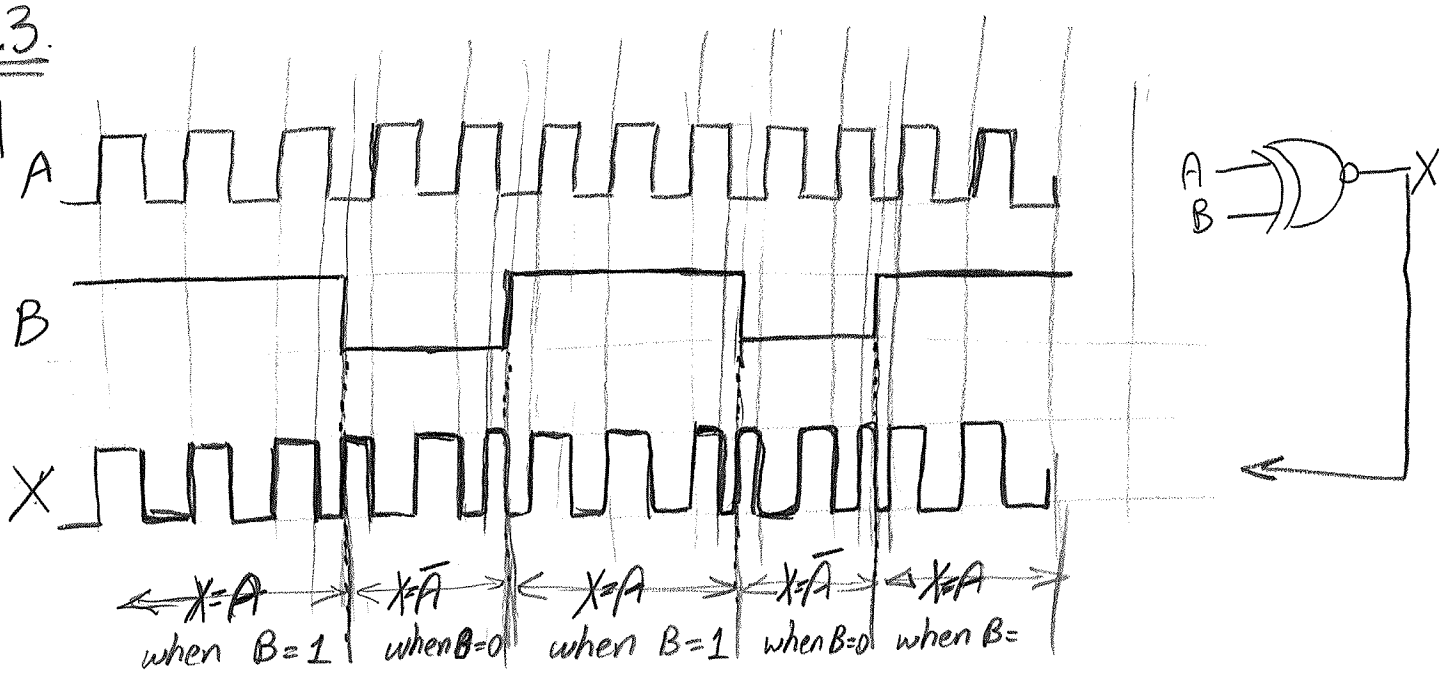
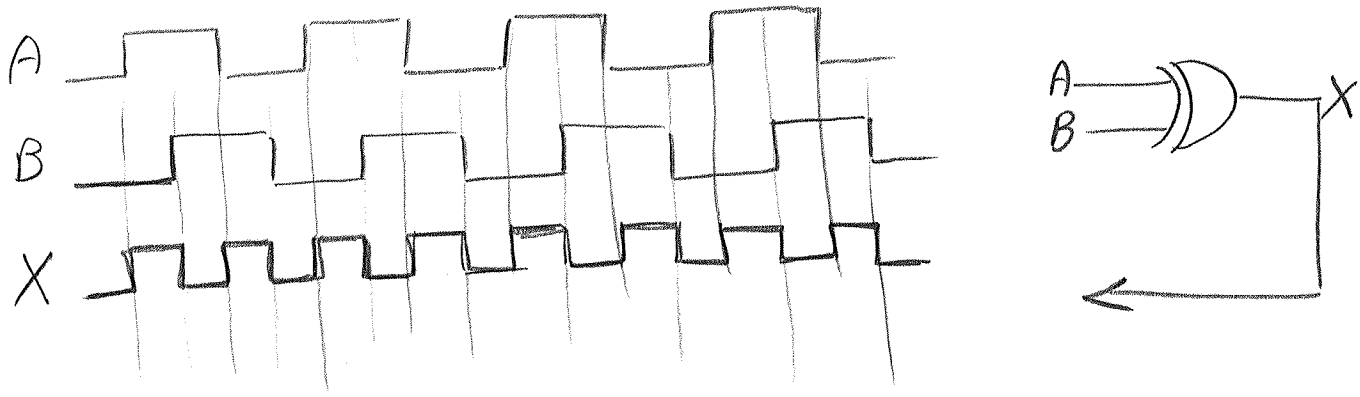

A
B
X
A
B
X
X=A
when B=1
X=$\bar{A}$
when B=0
X=A
when B=1
X=$\bar{A}$
when B=0
X=A
when B=

26


A
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